Auteur: Raj Shah
Studierichting: Informatica
Oefeningenreeks 3B nr. 4.5

Bereken de n-de afgeleide van

$f(x) = ax^4 + bx^3 + cx^2 + dx + e \ (a, b, c, d, e \in \mathbb{R})$

- $n = 1$

  $f'(x) = 4ax^3 + 3bx^2 + 2cx + d$

- $n = 2$

  $f''(x) = 12ax^2 + 6bx + 2c$

- $n = 3$

  $f'''(x) = 24ax + 6b$

- $n = 4$

  $f^{iv}(x) = 24a$

- $n = 5$

  $f^{v}(x) = 0$

$\Rightarrow n \geq 5 : f^{(n)}(x) = 0$

References